

Esercizio 1. Decomporre in fratti semplici

$$\begin{aligned} W(s) &= \frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)} \\ &= \frac{s^2 + 2s + 3}{((s+1)^2 + 1)((s+1)^2 + 2^2)} \\ &= \frac{A(s+1) + B}{((s+1)^2 + 1^2)} + \frac{C(s+1) + D}{((s+1)^2 + 2^2)} \end{aligned}$$

$$\begin{aligned} s^2 + 2s + 2 &= (s+1)^2 + 1 \\ a &= \frac{2}{2} = 1 \\ b &= 2 - 1^2 = 1 \\ s^2 + 2s + 5 &= (s+1)^2 + 2^2 \end{aligned}$$

$$\begin{aligned} s^2 + \beta_1 s + \beta_2 &= (s+a)^2 + b^2 \\ &= s^2 + 2as + a^2 + b^2 \\ a &= \frac{\beta_1}{2} \\ a^2 + b^2 &= \beta_2 \\ b^2 &= \beta_2 - \left(\frac{\beta_1}{2}\right)^2 \end{aligned}$$

$$\frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)}((s+1)^2 + 1^2) = \frac{A(s+1) + B}{((s+1)^2 + 1^2)}((s+1)^2 + 1^2) + \frac{C(s+1) + D}{((s+1)^2 + 2^2)}((s+1)^2 + 1^2)$$

Scelgo s per cui $(s+1)^2 + 1 = 0 \Rightarrow s^2 = -2s - 2$

$$\begin{aligned} \frac{s^2 + 2s + 3}{s^2 + 2s + 5} &= A(s+1) + B \\ \frac{-2s - 2 + 3}{-2s - 2 + 5} &= A s + A + B \\ 0s + \frac{1}{3} &= A s + A + B \end{aligned}$$

$$\begin{aligned} A &= 0 \\ A + B &= \frac{1}{3} \\ B &= \frac{1}{3} \end{aligned}$$

$$\frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)}((s+1)^2 + 2^2) = \frac{A(s+1) + B}{((s+1)^2 + 1^2)}((s+1)^2 + 2^2) + C(s+1) + D$$

prendo s tale che $(s+1)^2 + 2^2 = 0 \Rightarrow s^2 = -2s - 5$

$$\begin{aligned} \frac{s^2 + 2s + 3}{s^2 + 2s + 2} &= C(s+1) + D \\ \frac{-2s - 5 + 3}{-2s - 5 + 2} &= C(s+1) + D \\ 0s + \frac{-2}{-3} &= C s + C + D \end{aligned}$$

$$\begin{aligned} C &= 0 \\ D &= \frac{2}{3} \end{aligned}$$

Quindi

$$W(s) = \frac{\frac{1}{3}}{((s+1)^2+1^2)} + \frac{\frac{2}{3}}{((s+1)^2+2^2)}$$

Vediamo

$$\begin{aligned} W(s) &= \frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)} \\ &= \frac{s^2+2s+2+1}{(s^2+2s+2)(s^2+2s+5)} \\ &= \frac{s^2+2s+2}{(s^2+2s+2)(s^2+2s+5)} + \frac{1}{(s^2+2s+2)(s^2+2s+5)} \\ &= \frac{1}{s^2+2s+5} + \frac{1}{(s^2+2s+2)(s^2+2s+5)} \end{aligned}$$

Proviamo metodo alternativo

$$\begin{aligned} \frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)} &= \frac{A(s+1)+B}{((s+1)^2+1^2)} + \frac{C(s+1)+D}{((s+1)^2+2^2)} \\ \lim_{s \rightarrow \infty} s \frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)} &= \lim_{s \rightarrow \infty} s \frac{A(s+1)+B}{((s+1)^2+1^2)} + \lim_{s \rightarrow \infty} s \frac{C(s+1)+D}{((s+1)^2+2^2)} \\ 0 &= A+C \\ C &= 0 \\ \frac{3}{10} &= \frac{B}{2} + \frac{D}{5} \\ 3 &= 5B+2D \\ 3-\frac{5}{3} &= 2D \\ D &= \frac{4}{6} = \frac{2}{3} \end{aligned}$$

Esercizio 2. Determinare un sistema che ha come soluzione generale

$$\begin{aligned} y(t) &= \delta_0(t-1)c_1 + \delta_0(t)c_2 \\ \delta_0(t) &= 0^t \text{ impulso unitario} \\ &= \begin{cases} 0 & \text{se } t \neq 0 \\ 1 & \text{se } t = 0 \end{cases} \\ \delta_0(t-1) &= 0^{t-1} \\ &= \begin{cases} 0 & \text{se } t \neq 1 \\ 1 & \text{se } t = 1 \end{cases} \\ y_0 &= y(t) \\ &= \delta_0(t-1)c_1 + \delta_0(t)c_2 \\ y_1 &= y(t+1) \\ &= \delta_0(t)c_1 + \underbrace{\delta_0(t+1)c_2}_{\text{sarebbe } \neq 0 \text{ solo a } t=-1, \text{ ma unilatera } \Rightarrow = 0} \\ &= \delta_0(t)c_1 \\ y_2 &= y(t+2) \\ &= \delta_0(t \neq 1)c_1 + \delta_0(t \neq 2)c_2 \\ &= 0 \end{aligned}$$

$$\begin{aligned}
y(t+2) &= 0y(t) + 0y(t+1) \\
x_1(t) &= y(t) \\
x_2(t) &= y(t+1)
\end{aligned}$$

$$\begin{aligned}
x_1(t+1) &= y(t+1) \\
&= x_2(t) \\
x_2(t+1) &= y(t+2) \\
&= 0 \\
A &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\
C &= \begin{pmatrix} 1 & 0 \end{pmatrix}
\end{aligned}$$

Esercizio 3. Calcolare un sistema che ha come soluzione generale

$$y(t) = e^t c_1 + e^{2t} c_2 + t e^t$$

$$\begin{aligned}
y_0 &= e^t c_1 + e^{2t} c_2 + t e^t \\
y_1 &= \text{diff}(e^t c_1 + e^{2t} c_2 + t e^t, t) \\
&= 2 c_2 e^{2t} + t e^t + c_1 e^t + e^t \\
y_2 &= \text{diff}(e^t c_1 + e^{2t} c_2 + t e^t, t, 2) \\
&= 4 c_2 e^{2t} + t e^t + c_1 e^t + 2 e^t
\end{aligned}$$

$$\begin{aligned}
\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} &= \begin{pmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} t e^t \\ t e^t + e^t \end{pmatrix} \\
\begin{pmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} &= \begin{pmatrix} y_0 \\ y_1 \end{pmatrix} - \begin{pmatrix} t e^t \\ t e^t + e^t \end{pmatrix} \\
\det \begin{pmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{pmatrix} &= e^{3t} \\
\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} &= \begin{pmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{pmatrix}^{-1} \cdot \begin{pmatrix} y_0 - t e^t \\ -(t e^t) - e^t + y_1 \end{pmatrix} \\
&= \frac{1}{e^{3t}} \begin{pmatrix} 2e^{2t} & -e^{2t} \\ -e^t & e^t \end{pmatrix} \cdot \begin{pmatrix} y_0 - t e^t \\ -(t e^t) - e^t + y_1 \end{pmatrix} \\
&= \begin{pmatrix} -(e^{-t} (t e^t - e^t + y_1 - 2 y_0)) \\ -(e^{-(2t)} (e^t - y_1 + y_0)) \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
y_2 &= 4 c_2 e^{2t} + t e^t + c_1 e^t + 2 e^t \\
&= 4 (-(e^{-(2t)} (e^t - y_1 + y_0))) e^{2t} + t e^t + (-(e^{-t} (t e^t - e^t + y_1 - 2 y_0))) e^t + 2 e^t \\
&= -(4 (e^t - y_1 + y_0)) + 3 e^t - y_1 + 2 y_0 \\
&= -e^t + 3 y_1 - 2 y_0
\end{aligned}$$

$$\ddot{y}(t) = 3 \dot{y}(t) - 2 y(t) - e^t$$

$$\begin{aligned}
x_1(t) &= y(t) \\
x_2(t) &= \dot{y}(t)
\end{aligned}$$

$$\begin{aligned}
\dot{x}_1(t) &= x_2(t) \\
\dot{x}_2(t) &= 3 \dot{y}(t) - 2 y(t) - e^t \\
&= 3 x_2(t) - 2 x_1(t) - e^t
\end{aligned}$$

$$\begin{aligned}
A &= \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \\
B &= \begin{pmatrix} 0 \\ -1 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
C &= \begin{pmatrix} 1 & 0 \end{pmatrix} \\
u(t) &= e^t \\
y(t) &= Cx(t) + Du(t) \\
&= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}
\end{aligned}$$

Esercizio 4. Trovare un sistema la cui soluzione generale è

$$y(t) = e^t \cos(3t) c_1 + e^t \sin(3t) c_2 + \cos(3t)$$

```

(%i1) Y:%e^t*cos(3*t)*c[1]+%e^t*sin(3*t)*c[2]+cos(3*t)
(%o1) c2 e^t sin(3 t) + c1 e^t cos(3 t) + cos(3 t)
(%i5) Y0:Y
(%o5) c2 e^t sin(3 t) + c1 e^t cos(3 t) + cos(3 t)
(%i6) Y1:diff(Y0,t)
(%o6) c2 e^t sin(3 t) - 3 c1 e^t sin(3 t) - 3 sin(3 t) + 3 c2 e^t cos(3 t) + c1 e^t cos(3 t)
(%i7) Y2:diff(Y1,t)
(%o7) -(8 c2 e^t sin(3 t)) - 6 c1 e^t sin(3 t) + 6 c2 e^t cos(3 t) - 8 c1 e^t cos(3 t) - 9 cos(3 t)
(%i8) sol:solve([y0-Y0,y1-Y1],[c[1],c[2]])
(%o8) [[c1 = (sin(3 t) (-y1 - cos(3 t)) + (sin(3 t) + 3 cos(3 t)) y0 - 3 sin(3 t)^2 - 3 cos(3 t)^2) / (3 e^t sin(3 t)^2 + 3 e^t cos(3 t)^2), c2 = (cos(3 t) y1 + (3 sin(3 t) - cos(3 t)) y0 + cos(3 t)^2) / (3 e^t sin(3 t)^2 + 3 e^t cos(3 t)^2)]]
(%i13) eq:factor(substitute(sol,y2-Y2))
(%o13) y2 - 2 y1 + 10 y0 - 6 sin(3 t) - cos(3 t)
(%i14)
(%i14)

```

$$\ddot{y}(t) - 2 \dot{y}(t) + 10 y(t) - 6 \sin(3t) - \cos(3t) = 0$$

$$\begin{aligned}
x_1(t) &= y(t) \\
x_2(t) &= \dot{y}(t)
\end{aligned}$$

$$\begin{aligned}
\dot{x}_1(t) &= \dot{y}(t) = x_2(t) \\
\dot{x}_2(t) &= \ddot{y}(t) \\
&= 2 \dot{y}(t) - 10 y(t) + 6 \sin(3t) + \cos(3t) \\
&= 2 x_2(t) - 10 x_1(t) + 6 \sin(3t) + \cos(3t) \\
A &= \begin{pmatrix} 0 & 1 \\ -10 & 2 \end{pmatrix} \\
B &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
u(t) &= 6 \sin(3t) + \cos(3t) \\
C &= \begin{pmatrix} 1 & 0 \end{pmatrix}
\end{aligned}$$

Esercizio 5. Trovare un sistema la cui soluzione generale è

$$y(t) = (-1)^t (c_1 + t c_2) + t$$

```
(%i1) Y:(-1)^t*(c[1]+t*c[2])+t
(%o1) (c2*t+c1)*(-1)^t+t
(%i2) Y0:Y
(%o2) (c2*t+c1)*(-1)^t+t
(%i4) Y1:factor(subst(t=t+1,Y0))
(%o4) -(c2*t*(-1)^t+c2*(-1)^t+c1*(-1)^t-t-1)
(%i5) Y2:factor(subst(t=t+2,Y0))
(%o5) c2*t*(-1)^t+2*c2*(-1)^t+c1*(-1)^t+t+2
(%i6) sol:solve([y0-Y0,y1-Y1],[c[1],c[2]])
(%o6) [[c1 = (t*(y1-2)+(t+1)*y0-2*t^2)/(-1)^t, c2 = -(y1+y0-2*t-1)/(-1)^t]]
(%i8) factor(subst(sol,y2-Y2))
(%o8) y2+2*y1+y0-4*t-4
(%i9)
```

$$\begin{aligned} y(t+2) + 2y(t+1) + y(t) - 4t - 4 &= 0 \\ x_1(t) &= y(t) \\ x_2(t) &= y(t+1) \end{aligned}$$

$$\begin{aligned} x_1(t+1) &= x_2(t) \\ x_2(t+1) &= y(t+2) \\ &= -2y(t+1) - y(t) + 4t + 4 \\ &= -2x_2(t) - x_1(t) + 4t + 4 \end{aligned}$$

$$\begin{aligned} A &= \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} \\ B &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ u(t) &= 4t + 4 \\ C &= (1 \ 0) \end{aligned}$$